

PACER: A Perturbation-Aware Contrastive Framework for Long-Tail Multimodal Recommendation

Minh-Khoi Le*

VNU University of Engineering and Technology, Hanoi, Vietnam
khoilm@vnu.edu.vn

Abstract—Multimodal recommenders regularise graph propagation with an InfoNCE contrastive head, yet the frozen ResNet-50 and Sentence-BERT features they consume carry sensor and semantic noise that content-graph propagation compounds—penalising long-tail items, whose small neighbourhoods average that noise away least effectively. We propose PACER (Perturbation-Aware Contrastive Encoder Regularisation), whose central finding is *distributional*: a lightweight two-view denoiser — DVR (Dual-View Reweighting) — reallocates capacity from Head to Mid/Tail at nearly constant overall Recall@20. DVR learns semantic-aware (SAV) and interest-aware (IAV) views, fuses them by a complement-weighted consensus, and pulls the trunk toward the denoised view via InfoNCE. In a matched A0-vs.-A1 ablation that toggles DVR alone, Mid- and Tail-tercile Recall@20 rise by +6.1%/+12.1% on Amazon Clothing and +38.1%/+65.3% on Amazon Sports ($d_z=4.32/5.52$, $p<10^{-3}$), with Head statistically flat on both and overall Recall@20 nearly unchanged. Against a capacity-matched MMHCL baseline — the same code with the embedding width raised to PACER’s $d=320$ — PACER gains +10.48% Recall@20 and +13.07% NDCG@20 on Clothing (+7.04%/+8.56% on Sports); the larger gains against MMHCL’s native $d=64$ setting (+18.92%/+21.17%) are substantially attributable to embedding capacity alone. All of this costs only $d+2$ extra parameters (322 at $d=320$).

Index Terms—Multimodal recommendation, graph contrastive learning, view perturbation, noise-robust item modeling, long-tail recommendation.

I. INTRODUCTION

Multimodal recommender systems fuse image, text, and interaction signals over the user–item graph, and now form the backbone of e-commerce, streaming, and short-video platforms [1]–[3]. Recent work has pushed representation quality with hypergraph propagation [1], [3], spectrum-based fusion [4], [5], dual-path modality routing [6], interest-tree augmentation of the modality graph [7], noise-robust item modelling with dynamic multi-view contrastive learning [8], and multi-level self-supervision [9]. Regardless of fusion strategy, however, the training signal that ties everything together is almost invariably the InfoNCE contrastive loss.

The failure mode we target is upstream of the loss: the frozen encoders are noisy, and propagation amplifies rather than attenuates that noise. Background clutter in images and boilerplate in product titles enter as sensor and semantic noise;

content-graph propagation then compounds them across layers before content is fused with the collaborative representation. Long-tail items are the primary casualties: their neighbourhoods are small, so the neighbourhood averaging that graph propagation relies on cannot smooth out the noise, and the InfoNCE head then absorbs the resulting representation drift into its logits. By *perturbation* we mean exactly this propagated multimodal noise, not an adversarial attack.

This long-tail harm is not only a technical pathology but the research target. Long-tail items form the bulk of the catalogue yet receive little exposure; improving Mid/Tail retrieval raises coverage, diversity, and long-run marketplace revenue. Aggregate top- K metrics, which current multimodal papers predominantly report, can mask the fact that gains accrue mainly to the Head—precisely the gap that our Head/Mid/Tail tercile evaluation is designed to expose.

Our contributions.

- *Distributional finding.* On two Amazon catalogues, adding a $d+2$ -parameter two-view denoiser to an MMHCL-style trunk reallocates representational capacity from Head to Mid/Tail at nearly constant overall Recall@20. In the matched A0-vs.-A1 pipeline (same width, depth, and schedule) DVR adds +6.1% Mid and +12.1% Tail on Clothing and +38.1% Mid and +65.3% Tail on Sports, with Head flat on both; against a capacity-matched MMHCL baseline the 250-epoch cross-model gains are +24.5% Mid and +20.8% Tail on Clothing (+45.7%/+46.5% on Sports).
- *Parameter-efficient architecture.* The DVR module (Dual-View Reweighting) is a drop-in view-contrastive denoiser that composes a semantic-aware view (SAV) and an interest-aware view (IAV) and fuses them by *subtracting* the softmax-weighted mixture from the raw sum (Eq. 4) — a complement-weighted consensus rule that upweights edges on which both views agree. DVR adapts the two-view design of NRDMC [8] but drops its prototype-aware view and its M learnable prototypes, costing $d+2 = 322$ extra parameters at the reported $d=320$ on both datasets.
- *Transparent negative-result audit.* A 2×2 factorial (Sec. V-E) over an RSFP-Growth-blended TAMER-style interest tree and a LogQ-Laplace popularity debias shows that neither axis contributes measurably once DVR is in place, and the

*Corresponding author.

only paired contrast reaching $\alpha=0.05$ favours *disabling* both (B11 vs. B00, $p=0.040$).

Provenance and headline numbers. PACER reimplements MACP whitening and the interest-tree item-item construction following TAMER [7]; Sec. IV-D states what we adopt, modify, and drop. On Amazon Clothing over five fixed seeds, PACER lifts Recall@20 by +10.48%, NDCG@20 by +13.07%, and Precision@20 by +10.28% over a capacity-matched MMHCL reproduction at the same embedding width (Table I also reports the larger figures against MMHCL’s native $d=64$ setting). The Sports replication (Sec. V-F) reproduces the tercile signature at higher statistical power.

II. RELATED WORK

Multimodal graph recommenders and noise-robust modelling. Modern multimodal recommenders combine collaborative propagation with modality-specific or fused content-graph branches and regularise them with contrastive objectives. MMHCL [1] treats each modality as a hyper-edge; ITCoHD-MRec [3] adds cooperative hypergraph diffusion; spectrum-based fusion [4], [5], multi-view alignment [2], dual-path routing [6], MENTOR [9], and self-supervised multimodal GCNs [10] vary fusion and self-supervision schedules but retain vanilla InfoNCE. PACER is orthogonal: it augments the encoder with a two-view denoising branch without changing modality fusion or the primary loss family. Contrastive-SSL foundations [11]–[13] justify InfoNCE-style regularisation; sampled-metric analysis [14] motivates our long-tail tercile evaluation. NRDMC [8] couples noise-robust item modelling with dynamic multi-view contrastive learning over three views (semantic, interest, prototype-aware); DVR adapts its two-view design to a parameter-efficient regime, retaining semantic-aware and interest-aware views with adaptive edge-weight fusion and a view-level LightGCN, but dropping the prototype-aware view and its M learnable prototypes $P \in \mathbb{R}^{M \times d}$.

Interest-tree augmentation and popularity debiasing. TAMER [7] redistributes modality features via a Multi-component Analysis Collaborative Processing (MACP) block (PCA→ICA and ZCA whitening) and augments the item-item modality graph with an Interest Tree built by BFS over co-occurrence. PACER reimplements these steps following TAMER; reported runs use the PCA→ICA replacement stream (Sec. IV-D); optionally we source the interest graph from an RSFPGrowth-blended co-occurrence and wire the augmented similarity into DVR’s item-side content graph. Log Q -corrected softmax is a standard InfoNCE popularity-bias remedy. Sec. V-E audits both axes and reports them as neutral/negative once DVR is present.

Denoising-diffusion recommenders. Recent denoising-diffusion work [15]–[17] addresses noise at the generative layer rather than at the encoder or the loss; DVR composes freely with any of these directions.

III. PRELIMINARIES

Let $\mathcal{U}$ be the user set, $\mathcal{I}$ the item set, and $\mathcal{G} = (\mathcal{U} \cup \mathcal{I}, \mathcal{E})$ the observed bipartite interaction graph. Each item $i \in \mathcal{I}$

carries a frozen visual feature $v_i \in \mathbb{R}^{d_v}$ (ResNet-50 pooled), a frozen textual feature $t_i \in \mathbb{R}^{d_t}$ (Sentence-BERT), and trainable ID embeddings. The goal is to learn a scoring function $s_\theta(u, i)$ that maximises Recall@ K and NDCG@ K on the held-out test set. Following LightGCN, PACER’s collaborative trunk propagates ID ego embeddings over L layers of the symmetrically normalised adjacency $\tilde{A}$:

$$Z^{(\ell+1)} = \tilde{A} Z^{(\ell)}, \quad Z = \frac{1}{L+1} \sum_{\ell=0}^L Z^{(\ell)}. \quad (1)$$

Standard BPR ranking supplies the primary loss $\mathcal{L}_{\text{BPR}}$; DVR (Sec. IV-B) contributes an auxiliary in-batch-negative InfoNCE view loss $\mathcal{L}_{\text{view}}$.

IV. THE PACER FRAMEWORK

A. Overview

Figure 1 summarises PACER end to end.

a) *Mechanical feature path:* Let $\tilde{v}_i, \tilde{t}_i$ be the frozen image and text features after the selected offline MACP transform. Each modality $m \in \{v, t\}$ is linearly projected to $\mathbb{R}^d$ and passed through a gated residual calibration route, giving $r_i^v, r_i^t \in \mathbb{R}^d$.¹ With trainable collaborative ID lookups q_u, q_i , content-branch lookups c_u, c_i , and the normalised user/item content adjacencies $\tilde{A}_U, \tilde{A}_I$ propagated for L_U, L_I steps, the actual fusion is

$$\begin{aligned} Z &= \text{LightGCN}_{\tilde{A}}([q_u; q_i]), & x_i &= \frac{1}{2}(r_i^v + r_i^t), \\ H_i &= \tilde{A}_I^{L_I}(c_i + x_i), & H_u &= \tilde{A}_U^{L_U}c_u, \\ E_i &= Z_i + \text{Norm}(H_i), & E_u &= Z_u + \text{Norm}(H_u). \end{aligned} \quad (2)$$

Thus the trunk ego embeddings are ID-only: image and text are neither concatenated with nor added to q_i before trunk LightGCN. They are mean-fused inside the parallel item-content branch, whose normalised output is added to the collaborative representation for BPR scoring. The user ego is a trainable lookup. This is early fusion within the content branch followed by late fusion between content and collaborative branches; it does not use TAMER’s ID-gated per-modality mixing or three-stream propagation.

DVR (Sec. IV-B) uses the collaborative anchors Z_u, Z_i and generates a two-view edge reweighting of the observed user-item graph and produces auxiliary view embeddings $\tilde{E}_u, \tilde{E}_i$ via a K -step view LightGCN. The BPR head provides pair-wise ranking on the final E_u, E_i , and an in-batch-negative InfoNCE view loss $\mathcal{L}_{\text{view}}$ pulls the view embedding toward Z_u, Z_i to encourage noise-robust representations. Both heads sum into the PACER objective (Sec. IV-C).

¹Projection $h_i^m = W_m \tilde{x}_i^m + b_m$; residual route $r_i^m = h_i^m + a_m \text{LN}(\mathcal{C}_m(h_i^m))$, where $\mathcal{C}_m$ is the enabled per-modality spectral calibration path in our released implementation (reported runs disable metadata-aware phase calibration and anti-variance filtering, retaining only the residual Inter-Modal Coherence Filter), LN is LayerNorm, and a_m is its learnable scalar residual-routing gate. The matched A1 control uses the same encoder path, so this route is held fixed in the DVR ablation.

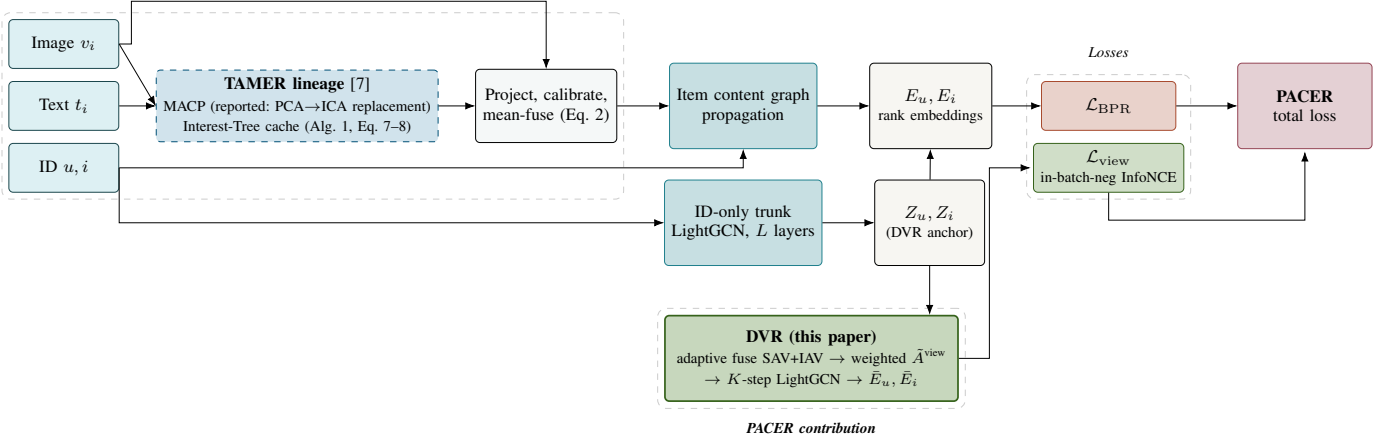


Figure 1. Overview of PACER. The collaborative trunk starts from user/item ID lookups; image and text follow a separate projected, calibrated item-content branch and are fused with the trunk only after propagation (Eq. 2). The novel **DVR** branch reweights the observed user-item graph from the collaborative anchors: its two-view (SAV + IAV) reweighting is propagated for K LightGCN steps to produce view embeddings $\bar{E}_u, \bar{E}_i$; an in-batch-negative InfoNCE view loss $\mathcal{L}_{\text{view}}$ pulls Z_u, Z_i toward the view embedding. The blue dashed box on the encoder side is the **TAMER lineage** [7]: MACP preprocessing (reported: PCA $\rightarrow$ ICA replacement) and the Interest-Tree cache (BFS Alg. 1 with Eq. 7–8); Section IV-D states what we reimplement, modify, and drop from TAMER.

B. DVR: A Dual-View Reweighting Denoiser

Motivation. Frozen image and text encoders inject sensor and semantic noise. Content-graph propagation can compound this perturbation before content is fused with the collaborative representation, disproportionately harming long-tail items whose sparse evidence offers weaker averaging. Following the noise-robust item modelling design of NRDMC [8], we generate a second, edge-reweighted view of the user-item graph and pull the trunk representation toward it via an InfoNCE loss. Unlike NRDMC’s three-view design, DVR uses only two views (SAV and IAV)—the popularity-aware prototype view is dropped—yielding a substantially lighter module that composes with any LightGCN-style backbone.

Two views. Given the L_2 -normalised collaborative anchors $\hat{Z}_u, \hat{Z}_i$, DVR scores every observed edge (u, i) under two complementary views. The *semantic-aware view* (SAV) uses cosine similarity, while the *interest-aware view* (IAV) uses a shared learnable attention vector $g \in \mathbb{R}^d$ and softmax-normalises within each user’s neighbourhood $\mathcal{N}(u)$:

$$s_{ui}^{\text{sav}} = \sigma(\hat{Z}_u^\top \hat{Z}_i), \quad s_{ui}^{\text{iav}} = \frac{\exp(\sigma(\hat{Z}_u^\top g)(\hat{Z}_i^\top g))}{\sum_{j \in \mathcal{N}(u)} \exp(\sigma(\hat{Z}_u^\top g)(\hat{Z}_j^\top g))}. \quad (3)$$

Here $\sigma(\cdot)$ denotes the logistic sigmoid $\sigma(x) = 1/(1 + e^{-x})$, which maps trunk inner products into $(0, 1)$ before they enter the softmax; $g \in \mathbb{R}^d$ is the sole per-view learnable parameter of DVR (beyond the two scalars introduced below).

What the shared vector learns. Initialised as $g \sim \mathcal{N}(0, I/d)$, the vector is a catalogue-level compatibility direction, not a standalone representation of every user’s interests. Equivalently, $(\hat{Z}_u^\top g)(\hat{Z}_i^\top g) = \hat{Z}_u^\top (g g^\top) \hat{Z}_i$, so IAV is a rank-one shared bilinear compatibility function learned jointly with the recommendation objective. Personalisation is supplied by the user anchor $\hat{Z}_u$, the item anchors, and normalisation over the

user-specific neighbourhood $\mathcal{N}(u)$. Hence two users generally induce different IAV distributions despite sharing g . A single direction deliberately preserves the $d+2$ parameter budget; richer user-conditioned or multi-head probes are left for future work.

Adaptive fusion and view LightGCN. A scalar affine $(W_{\text{fuse}}, b_{\text{fuse}}) \in \mathbb{R}^2$ produces per-view logits $f_{ui}^v = \tanh(W_{\text{fuse}} s_{ui}^v + b_{\text{fuse}})$; a softmax over $v \in \{\text{sav}, \text{iav}\}$ yields adaptive mixing weights $\alpha_{ui}^v = \text{softmax}_v(f_{ui}^v)$, and the fused edge weight is

$$w_{ui}^{\text{final}} = s_{ui}^{\text{sav}} + s_{ui}^{\text{iav}} - (\alpha_{ui}^{\text{sav}} s_{ui}^{\text{sav}} + \alpha_{ui}^{\text{iav}} s_{ui}^{\text{iav}}). \quad (4)$$

Intuition. Since $\alpha_{ui}^{\text{sav}} + \alpha_{ui}^{\text{iav}} = 1$, Eq. (4) algebraically rearranges to $(1 - \alpha_{ui}^{\text{sav}}) s_{ui}^{\text{sav}} + (1 - \alpha_{ui}^{\text{iav}}) s_{ui}^{\text{iav}}$: each view is weighted by the complement of its softmax weight, so edges on which SAV and IAV agree ($\alpha^v \approx \frac{1}{2}$) receive maximal fused weight while single-view-dominated edges are attenuated — an explicit consensus criterion.

The symmetrically normalised weighted adjacency $\hat{A}^{\text{view}}$ (edge values w_{ui}^{final}) is propagated for K LightGCN steps, yielding L_2 -normalised view embeddings $\bar{E}_u, \bar{E}_i$ that enter the view loss below. Trunk embeddings are not replaced by these view embeddings: DVR acts as an auxiliary view generator whose contrastive pull regularises the collaborative trunk, while BPR uses the final content-collaborative fusion in Eq. (2).

Parameter cost. DVR adds only $g \in \mathbb{R}^d$ and the two scalars $(W_{\text{fuse}}, b_{\text{fuse}})$: $d+2$ extra parameters (322 on each dataset), three orders of magnitude below a Linear-GELU-Linear residual block—so the module is parameter-efficient by construction.

C. Total Objective

PACER retains the MMHCL item/user contrastive terms and adds the DVR view loss to BPR ranking and ℓ_2 regularisation:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{BPR}} + \lambda_i \mathcal{L}_{\text{item}} + \lambda_u \mathcal{L}_{\text{user}} + \lambda_{\text{cl}} \mathcal{L}_{\text{view}} + \lambda_{\text{reg}} \|\theta\|_2^2. \quad (5)$$

The view term is a symmetric in-batch-negative InfoNCE between the normalised collaborative anchor $\hat{Z}$ and the DVR view $\bar{E}$, sharing the InfoNCE temperature τ :

$$\mathcal{L}_{\text{view}} = \frac{1}{2}[\ell_{\text{NCE}}(\hat{z}, \bar{z}; \tau) + \ell_{\text{NCE}}(\bar{z}, \hat{z}; \tau)], \quad (6)$$

where $\hat{z}, \bar{z}$ are the L_2 -normalised anchor and view embeddings and ℓ_{NCE} is the standard batch- B InfoNCE with cosine similarity divided by τ . Eq. (6) is applied independently on the user and item branches and averaged. The loss weights, temperature, and view depth are dataset-specific; their reported values appear in Sec. V-A.

D. Relation to TAMER: What We Reuse, Modify, and Drop

The encoder-side preprocessing and the item-item interest-tree cache are reimplemented following TAMER [7]: we use MACP’s PCA→ICA replacement and the Interest-Tree BFS construction (Alg. 1, Eq. 7–8); we change the interest-graph source (optional RSFPGrowth blend, audited in Sec. V-E) and integrate the resulting adjacency into PACER’s item-content branch; we do not use TAMER’s three-stream graph learning, ID-gated modality purification, per-modality mixing, or BPR-only loss.² Sec. V-E audits the optional RSFPGrowth source blend: the reported model disables this blend but retains the base co-occurrence Interest-Tree.

V. EXPERIMENTS

We organise the empirical study around five questions. RQ1–RQ4 are centred on Amazon Clothing; RQ1 and RQ5 also report Amazon Sports (Optuna-retuned PACER; five-seed MMHCL baseline on the same fixed seeds).

- **RQ1.** Does PACER improve top- K accuracy over a strong multimodal contrastive baseline, and are the gains stable across seeds?
- **RQ2.** Is DVR individually necessary?
- **RQ3.** Do the gains reach the long-tail items?
- **RQ4.** Do the auxiliary Data (TAMER-style Interest-Tree with RSFPGrowth-blended source) and Loss (LogQ-Laplace) axes contribute? (Section V-E.)
- **RQ5.** Does the distributional effect of DVR replicate on a second dataset (Amazon Sports)? (Section V-F.)

A. Setup

Datasets. We report on Amazon Clothing 5-core ($|\mathcal{U}| \approx 39\text{K}$, $|\mathcal{I}| = 23,033$, $\approx 278\text{K}$ interactions) and Amazon Sports 5-core ($|\mathcal{U}| \approx 35\text{K}$, $|\mathcal{I}| = 18,357$, $\approx 256\text{K}$ interactions), each with a per-user 8:1:1 chronological split (integer rounding). Visual features come from ResNet-50 pooled activations and textual features from Sentence-BERT.

Baseline. We compare against MMHCL [1], the multimodal hyper-graph contrastive recommender in its native encoder configuration, without DVR. The MMHCL cells throughout Section V are our own five-seed reproduction on the same data

²Implementation details, including the MACP block and the Interest-Tree BFS with pruning $\lfloor |L_{o-1}| / ((o-1) \cdot 2) \rfloor$, follow Alg. 1 and Eq. 7–8 of [7]; our released code documents the correspondence.

split, frozen input features, and fixed seed set for both Clothing and Sports, *not* the published numbers of [1, Table 4].

Protocol. The five fixed random seeds³ are run on identical hardware (single RTX 5090). PACER training uses AdamW with a 100-epoch budget, early-stopping patience of 5 validation evaluations after a min_epochs=30 warm-up, and validation Recall@20 as the sole model-selection criterion. The reported PACER configuration uses $d=320$ and $L=3$ on both datasets. Clothing uses $B=1024$, $(\lambda_i, \lambda_u)=(0.07, 0.03)$, $\lambda_{\text{cl}}=0.10$, $\lambda_{\text{reg}}=4.8 \times 10^{-4}$, $\tau=0.30$, and DVR depth $K=2$. The independently Optuna-tuned Sports run uses $B=2048$, $(\lambda_i, \lambda_u)=(0.0197, 0.0123)$, $\lambda_{\text{cl}}=0.10$, $\lambda_{\text{reg}}=1.20 \times 10^{-3}$, $\tau=0.113$, and $K=3$. The auxiliary RSFPGrowth-blend and LogQ axes are disabled, while the base co-occurrence Interest-Tree is retained (Sec. V-E).

Scope of comparison. Table I deliberately reports one strong local baseline rather than a leaderboard-style table; a broader comparison against MENTOR, ITCohD-MRec, and spectrum-based fusion is a natural next step but beyond the scope of this paper.

Fairness. PACER and MMHCL share the chronological 8:1:1 split, frozen modality features, fixed seeds, and single-RTX 5090 hardware. Because MMHCL’s published configuration uses $d=64$ while PACER uses $d=320$, we re-ran the MMHCL code with only the embedding width changed to 320, holding every other flag, the split, and the five seeds fixed. Capacity alone lifts MMHCL Clothing Recall@20 from 0.08737 to 0.09405 (+7.6%), and lifts its Mid/Tail terciles substantially (Table III), so we report both baselines and quote Δ_{320} throughout. Residual depth ($L=3$ vs. $L=2$) and training-protocol (100-epoch AdamW vs. 250-epoch Adam) differences are not controlled, so no cross-model row is by itself causal evidence for DVR; the ablation in Table II, which holds $d=320$, $L=3$, and every other setting fixed across A0/A1, is the primary evidence.

All PACER cells in Tables I–III share one A0 block, so every t -test is paired.

B. RQ1 — Top- K Accuracy

Table I reports the head-to-head comparison on Amazon Clothing over the five fixed random seeds and the matching replication on Amazon Sports. Against the capacity-matched MMHCL baseline, PACER lifts Clothing Recall@20 from 0.09405 to 0.10390 (+10.48%), NDCG@20 from 0.04187 to 0.04734 (+13.07%), and Precision@20 from 0.004769 to 0.005260 (+10.28%); a little over half of the nominal $d=64$ margin survives once embedding width is equalised. Because both models are run on the same five seeds we use a paired t -test ($\text{df}=4$, Bonferroni $m=3$), which gives $t \geq 21.4$ and $p_{\text{Bonf}} < 10^{-4}$ on all three metrics, with PACER ahead on 5 of 5 seeds ($d_z \geq 9.6$). On Sports the same pipeline (Optuna-retuned; Sec. V-F) lifts Recall/NDCG/Precision@20 by +7.04/+8.56/+7.15% over the capacity-matched baseline, against +14.98/+17.28/+15.18% at $d=64$.

³Seeds: 1,616,406,634, 1,640,104,851, 52,093,548, 109,649,638, 372,270,914; drawn once and reused across all variants in this paper.

Table I

RQ1 — TOP-20 ACCURACY, MEAN $\pm$ STD OVER THE FIVE FIXED RANDOM SEEDS. MMHCL IS OUR OWN REPRODUCTION ON THE SAME PREPROCESSING AND SEED SET, REPORTED AT ITS NATIVE $d=64$ AND AT A CAPACITY-MATCHED $d=320$ OBTAINED BY CHANGING ONLY THE EMBEDDING WIDTH (SEC. V-A); PACER USES $d=320$, SO Δ_{320} IS THE CONFOUND-REDUCED CONTRAST AND Δ_{64} IS FOR REFERENCE. SPORTS PACER IS OPTUNA-TUNED (100-EP). t IS A PAIRED t -TEST OVER THE MATCHED SEEDS (DF= 4), BONFERRONI-CORRECTED FOR $m=3$, AGAINST THE $d=320$ BASELINE.

Dataset	Metric	MMHCL $d=64$	MMHCL $d=320$	PACER (ours)	Δ_{320} (Δ_{64})	t (p_{Bonf})
Clothing	R@20	0.08737 $\pm$ 0.00073	0.09405 $\pm$ 0.00043	0.10390$\pm$0.00078	+10.48% (+18.92%)	21.4 (8.4×10^{-5})
	P@20	0.004426 $\pm$ 0.000037	0.004769 $\pm$ 0.000019	0.005260$\pm$0.000039	+10.28% (+18.84%)	21.4 (8.5×10^{-5})
	N@20	0.03907 $\pm$ 0.00034	0.04187 $\pm$ 0.00031	0.04734$\pm$0.00018	+13.07% (+21.17%)	26.9 (3.4×10^{-5})
Sports	R@20	0.10503 $\pm$ 0.00068	0.11283 $\pm$ 0.00056	0.12077$\pm$0.00058	+7.04% (+14.98%)	19.3 (1.3×10^{-4})
	P@20	0.005531 $\pm$ 0.000035	0.005946 $\pm$ 0.000026	0.006371$\pm$0.000030	+7.15% (+15.18%)	23.2 (6.2×10^{-5})
	N@20	0.04977 $\pm$ 0.00018	0.05377 $\pm$ 0.00014	0.05837$\pm$0.00024	+8.56% (+17.28%)	31.8 (1.7×10^{-5})

C. RQ2 — DVR Necessity

Table II consolidates the DVR leave-one-out ablation: Panel A on Clothing (this section), Panel B on Sports (Sec. V-F). We toggle DVR by setting $\lambda_{\text{cl}}=0$, so Eq. (6) is dropped from Eq. (5); all other hyperparameters stay at each dataset’s reported configuration. Seeds are matched across A0 and A1 and we report a paired- t contrast with $\text{df}=4$, the paired Cohen’s d_z , and the number of seeds on which A0 beats A1 (“dir”).

NDCG@20 rises significantly ($\Delta=+1.8\%$, $p=0.029$, $d_z=+1.48$, dir 5 of 5); Recall@20 and Precision@20 rise directionally with large paired effect sizes ($d_z=+0.89/+0.90$; dir 4 of 5), reaching strict significance only in the higher-powered Sports replication (Panel B, Sec. V-F). Head is flat within the matched A0-vs.-A1 pipeline, while Mid/Tail Recall@20 rise by +0.236 pp ($d_z=+0.69$) and +0.190 pp ($d_z=+0.82$), consistent with the design intent of DVR (cf. Table III, row Δ_{DVR}). Extended training to 250 epochs sharpens this trade-off (Sec. V-D).

Table II

DVR NECESSITY ACROSS DATASETS (FIVE FIXED SEEDS, 100-EP; MEAN $\pm$ STD $\times 100$). **PANEL A** AMAZON CLOTHING (RQ2). **PANEL B** AMAZON SPORTS (RQ5, OPTUNA-TUNED). A0 IS THE REPORTED PACER; A1 DISABLES DVR ($\lambda_{\text{cl}}=0$); STATISTICAL ROWS ARE PAIRED- t , DF=4; DIR COUNTS SEEDS ON WHICH A0 BEATS A1.

Panel A. Amazon Clothing (RQ2)			
Variant	R@20	N@20	P@20
A0 PACER-full	10.390$\pm$0.078	4.734$\pm$0.018	0.526$\pm$0.004
A1 – no DVR	10.276 $\pm$ 0.062	4.651 $\pm$ 0.049	0.520 $\pm$ 0.003
Δ (A0–A1)	+1.1%	+1.8%	+1.1%
$t/p/d_z$ / dir	1.99/0.117/0.89 4 of 5	3.32/0.029/1.48 5 of 5	2.02/0.113/0.90 4 of 5
Panel B. Amazon Sports (RQ5, Optuna-tuned)			
Variant	R@20	N@20	P@20
A0 PACER-full	12.077$\pm$0.058	5.837$\pm$0.024	0.637$\pm$0.003
A1 – no DVR	11.857 $\pm$ 0.118	5.678 $\pm$ 0.047	0.624 $\pm$ 0.006
Δ (A0–A1)	+1.9%	+2.8%	+2.1%
$t/p/d_z$ / dir	3.95/0.017/1.77 5 of 5	7.05/0.002/3.15 5 of 5	4.54/0.010/2.03 5 of 5

D. RQ3 — Long-Tail Coverage and the 250-Epoch Schedule

Items are partitioned into three popularity terciles by training-set frequency [14]; DVR should benefit long-tail items most because their small neighbourhoods amplify the propagated noise that the view-contrastive loss suppresses. Table III and Fig. 2 compare PACER at 100/250 epochs, its matched no-DVR ablation (A1), and MMHCL. At 100 epochs PACER im-

proves Mid/Tail by +13.8%/+4.3% over the capacity-matched baseline against a Head gain of +10.3%, and the matched A0-vs.-A1 contrast attributes +6.1%/+12.1% of that Mid/Tail level to DVR alone. At 250 epochs the cross-model gains reach +24.5%/+20.8% (paired against MMHCL₃₂₀ on the shared seeds: $p=0.017/0.030$, dir 5 of 5 and 4 of 5) while Head recedes to +8.7% and overall Recall@20 is unchanged ($\Delta=-0.002$ pp, $p=0.90$). The tercile exchange is itself significant (paired- t , $\text{df}=4$): Head $\Delta=-0.229$ pp ($p=0.042$, $d_z=-1.32$, dir 0 of 5), Mid $\Delta=+0.390$ pp ($p=0.025$, $d_z=+1.57$, dir 5 of 5), Tail $\Delta=+0.278$ pp ($p=0.037$, $d_z=+1.38$, dir 4 of 5).

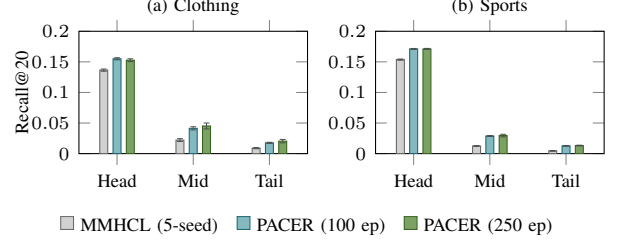


Figure 2. RQ3/RQ5 — Recall@20 by popularity tercile, five-seed mean $\pm$ std from Table III. A common y -axis scale is held fixed so the two catalogues remain comparable. DVR lifts Mid/Tail on both at 100 ep and reallocates further from Head at 250 ep; the effect is stronger on Sports. MMHCL bars use native $d=64$; see Table III for the capacity-matched control.

Table III

RQ3/RQ5 — RECALL@20 BY ITEM-POPULARITY TERCILE (EQUAL-SIZED UP TO INTEGER ROUNDING: CLOTHING 7,678/7,678/7,677; SPORTS 6,119 EACH). FIVE FIXED SEEDS THROUGHOUT. A1 IS THE DVR LEAVE-ONE-OUT ABLATION ($\lambda_{\text{cl}}=0$); Δ_{DVR} IS THE PAIRED A0–A1 CONTRAST. HEAD REFERS TO A0-VS.-A1 WITHIN THE MATCHED PIPELINE (STATISTICALLY FLAT), NOT TO ANY CROSS-MODEL COMPARISON. CROSS-MODEL $\Delta\%$ ROWS ARE QUOTED AGAINST THE CAPACITY-MATCHED MMHCL₃₂₀ CONTROL; ALL VALUES ARE TEST-SIDE. BOLDFACE HIGHLIGHTS THE MID/TAIL CELLS OF THE Δ_{DVR} ROW — THE PRIMARY LONG-TAIL EVIDENCE.

Panel A. Amazon Clothing				
Model	R@20 (all)	R@20 (Head)	R@20 (Mid)	R@20 (Tail)
MMHCL, native $d=64$	0.0874 $\pm$ 0.0007	0.1365 $\pm$ 0.0018	0.0221 $\pm$ 0.0023	0.0089 $\pm$ 0.0009
MMHCL, capacity-matched $d=320$	0.0940 $\pm$ 0.0004	0.1404 $\pm$ 0.0024	0.0363 $\pm$ 0.0044	0.0169 $\pm$ 0.0014
PACER-full, 100-ep (A0)	0.1039 $\pm$ 0.0008	0.1550 $\pm$ 0.0017	0.0413 $\pm$ 0.0026	0.0176 $\pm$ 0.0009
A1 – w/o DVR, 100-ep	0.1028 $\pm$ 0.0006	0.1545 $\pm$ 0.0011	0.0389 $\pm$ 0.0027	0.0157 $\pm$ 0.0017
Δ_{DVR} (A0–A1)	+0.0011 (+1.1%)	+0.0004 (+0.3%)	+0.0024 (+6.1%)	+0.0019 (+12.1%)
PACER-full (250 ep)	0.1039 $\pm$ 0.0008	0.1527 $\pm$ 0.0023	0.0452 $\pm$ 0.0048	0.0204 $\pm$ 0.0028
$\Delta\%$ (250 ep) vs. MMHCL ₃₂₀	+10.5%	+8.7%	+24.5%	+20.8%
Panel B. Amazon Sports (Optuna-tuned)				
Model	R@20 (all)	R@20 (Head)	R@20 (Mid)	R@20 (Tail)
MMHCL, native $d=64$	0.1050 $\pm$ 0.0007	0.1537 $\pm$ 0.0009	0.0125 $\pm$ 0.0006	0.0045 $\pm$ 0.0006
MMHCL, capacity-matched $d=320$	0.1128 $\pm$ 0.0006	0.1624 $\pm$ 0.0008	0.0202 $\pm$ 0.0007	0.0090 $\pm$ 0.0002
PACER-full, 100-ep (A0)	0.1208 $\pm$ 0.0006	0.1712 $\pm$ 0.0007	0.0288 $\pm$ 0.0008	0.0127 $\pm$ 0.0006
A1 – w/o DVR, 100-ep	0.1186 $\pm$ 0.0012	0.1711 $\pm$ 0.0014	0.0208 $\pm$ 0.0014	0.0077 $\pm$ 0.0008
Δ_{DVR} (A0–A1)	+0.0022 (+1.9%)	+0.0002 (+0.1%)	+0.0079 (+38.1%)	+0.0050 (+65.3%)
PACER-full (250 ep)	0.1209 $\pm$ 0.0005	0.1712 $\pm$ 0.0008	0.0295 $\pm$ 0.0021	0.0132 $\pm$ 0.0006
$\Delta\%$ (250 ep) vs. MMHCL ₃₂₀	+7.2%	+5.4%	+45.7%	+46.5%

E. RQ4 — Auxiliary Axes Factorial Audit

We audit a Data axis (RSFPGrowth source blend for the TAMER Interest Tree [7]) and a Loss axis (LogQ-Laplace), with DVR enabled in all four cells (Table IV). Under paired t , disabling both beats enabling both by +0.049 pp (B11 vs. B00, $p=0.040$, dir 5 of 5); each single-axis contrast is non-significant (Data, B11 vs. B10, $p=0.074$; Loss, B11 vs. B01, $p=0.82$). Neither axis adds measurable value once DVR is present.

Table IV

RQ4 — 2×2 FACTORIAL AUDIT (CLOTHING, FIVE SEEDS, 100-EP, R@20 $\times 100$). ROWS: LOGQ-LAPLACE; COLUMNS: RSFPGROWTH BLEND α .

DVR IS ON AND $\tau=0.30$ IN ALL FOUR CELLS. LABELS FOLLOW THE INTERNAL RUN NAMING, WHERE 1 DENOTES THE DISABLED SETTING; B11 HAS BOTH AXES OFF (PACER’S CONFIGURATION; TABLES I–III REPORT AN INDEPENDENT RUN OF IT, INDISTINGUISHABLE AT PAIRED $p=0.52$), B00 BOTH ON.

Config	RSFP $\alpha=0.10$	RSFP $\alpha=0$
LogQ on	B00: 10.358 ± 0.069	B01: 10.400 ± 0.074
LogQ off	B10: 10.377 ± 0.058	B11: 10.407± 0.035

F. RQ5 — Between-Dataset Replication (Amazon Sports)

We repeat RQ2 on Sports after an independent Optuna sweep (Tables II–III, Fig. 2b). In the matched ablation DVR adds +38.1% Mid and +65.3% Tail ($d_z=4.32/5.52$, $p<10^{-3}$), with Head flat ($d_z=0.11$). The cross-dataset gap is a *steeper popularity gradient* (not raw density): under MMHCL’s native $d=64$ configuration the Head:Mid:Tail R@20 ratio is 12.3:1:0.36 on Sports vs. 6.2:1:0.40 on Clothing, so neighbourhood averaging fails first on thin neighbourhoods, where DVR’s SAV/IAV consensus supplies structure-aware signal — hence the $\approx 6 \times$ larger Sports Δ_{DVR} . The 250-epoch schedule sharpens the trade-off (Mid +45.7%, Tail +46.5% vs. MMHCL₃₂₀, paired $p \leq 3 \times 10^{-4}$, dir 5 of 5) at unchanged overall Recall@20, mirroring Sec. V-D.

VI. CONCLUSION

PACER’s finding is *distributional*: a $d+2$ -parameter two-view denoiser (DVR) reallocates capacity from Head to Mid/Tail at nearly constant overall Recall@20. In the matched A0-vs.-A1 pipeline DVR adds +6.1% Mid and +12.1% Tail on Clothing and +38.1% Mid and +65.3% Tail on Sports, with Head flat on both and overall Recall@20 statistically unchanged ($p=0.90$ at 250 epochs). Against a capacity-matched MMHCL baseline at the same embedding width, PACER lifts Clothing Recall@20 by +10.48% and NDCG@20 by +13.07% (+7.04%/+8.56% on Sports), with 250-epoch Mid/Tail gains of +24.5%/+20.8%, and a 2×2 audit found neither auxiliary axis helpful. Practically, DVR offers marketplaces a way to buy catalogue coverage without paying for it in aggregate accuracy.

Limitations. We evaluate on two Amazon catalogues and a single backbone family. Our capacity control shows embedding width alone explains much of the nominal cross-model margin, and depth and protocol differences remain uncontrolled, so Table I is context rather than attribution. IAV’s single shared g restricts compatibility to one rank-one latent direction; multi-head or user-conditioned probes may be more expressive. Future work composes DVR with denoising-diffusion negative sampling [17] and tests transfer to non-MMHCL backbones.

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